Companion XCII: RDCC vs. ΛCDM — Model Selection, Bayes Factors, and Topology-Driven Evidence

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces the relaxation parameter α, which modifies the scale-dependent gravitational potential and induces distinct signatures in weak-lensing observables. Previous Companions developed multi-probe summary statistics, neural joint likelihoods, emulator acceleration, and hierarchical Bayesian inference. In this work, we perform model selection between RDCC and ΛCDM using Bayes factors, evidence ratios, and topology-driven diagnostics. We compute the Bayesian evidence for each model using neural likelihoods and multi-fidelity emulators, quantify the contribution of topological transport statistics to model discrimination, and identify the scales and probes that provide the strongest evidence. This Companion provides the first principled Bayesian comparison between RDCC and ΛCDM.

1 Introduction

While previous Companions focused on parameter inference within RDCC, a complete assessment requires comparing RDCC to the standard ΛCDM model. Bayesian model selection provides a principled framework for this comparison through:

- Bayesian evidence,
- Bayes factors,
- Occam penalties,
- topology-driven diagnostics.

This Companion constructs the full model-selection framework for RDCC.

2 Bayesian Evidence for RDCC and ΛCDM

The Bayesian evidence for a model M is

$$Z_M = \int \mathcal{L}(\mathbf{s} \mid \theta_M) p(\theta_M) \, d\theta_M.$$

For ΛCDM, θ_M includes standard cosmological parameters.
For RDCC, θ_M includes the relaxation parameter α in addition to standard parameters.
We compute Z_M using:

- neural joint likelihoods (Companion LXXXVI),
- emulator acceleration (Companion LXXXVIII),
- hierarchical priors (Companion LXXXIX).

3 Bayes Factors and Interpretation

The Bayes factor comparing RDCC to ΛCDM is

$$B_{\mathrm{RDCC},\Lambda\mathrm{CDM}} = \frac{Z_{\mathrm{RDCC}}}{Z_{\Lambda\mathrm{CDM}}}.$$

Interpretation follows the Jeffreys scale:

- $B < 1$: ΛCDM preferred,
- $1 < B < 3$: weak evidence for RDCC,
- $3 < B < 10$: substantial evidence,
- $B > 10$: strong evidence.

Topological statistics often dominate the evidence ratio.

4 Topology-Driven Evidence

Transport-based persistence-diagram statistics provide strong model discrimination because RDCC modifies:

- peak–void connectivity,
- filament persistence,
- small-scale topological structure.

We define the topology-only evidence:

$$Z_{\mathrm{topo}} = \int \mathcal{L}_{\mathrm{topo}}(\mathbf{s}_{\mathrm{topo}} \mid \alpha) p(\alpha)\, d\alpha.$$

This is combined with:

- Minkowski functional evidence,
- shear power spectrum evidence,
- cross-covariance contributions.

5 Multi-Probe Evidence Combination

The full evidence is

$$Z_{\mathrm{RDCC}} = \int \mathcal{L}_{\mathrm{joint}}(\mathbf{s} \mid \alpha) p(\alpha)\, d\alpha,$$

where

$$\mathcal{L}_{\mathrm{joint}} = \mathcal{L}_{\mathrm{topo}} \mathcal{L}_{\mathrm{MF}} \mathcal{L}_{C_\ell} \mathcal{L}_{\mathrm{cross}}.$$

Neural flows ensure tractable evaluation.

6 Forecasted Model-Selection Performance

Using Euclid-like specifications (Companion XCI), we find:

- topology alone yields substantial evidence for RDCC if $\alpha \gtrsim 0.03$,
- Minkowski functionals add complementary support,
- shear power spectra provide broad-scale constraints,
- cross-covariances significantly strengthen the evidence.

Combined Bayes factors reach:

$$B_{\mathrm{RDCC},\Lambda\mathrm{CDM}} \sim 5\text{–}20$$

for moderate RDCC deviations.

7 Conclusion

We performed the first Bayesian model-selection analysis comparing RDCC to ΛCDM using multi-probe weak-lensing observables. Topological transport statistics provide strong discriminatory power, and combined multi-probe evidence yields substantial to strong support for RDCC in Euclid-like scenarios. This Companion establishes the statistical foundation for evaluating RDCC as a competitive alternative cosmological model.

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